

SETU Student Support Services



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Introduction

SETU Student Support Services (S4) is an integrated digital platform designed to enhance the overall student and staff experience within SETU.

The system brings together three services: prioritized parking, lecturer and module ratings, and a student-driven information exchange marketplace into a single application.

The motivation behind *S4* is to address common challenges faced by students and staff, such as difficulty securing parking, lack of transparent academic feedback, and limited access to reliable study resources.

By centralizing these services, the platform aims to improve convenience and collaboration across the college.

S4 is designed with simplicity, accessibility, and efficiency in mind. It incorporates features such as secure authentication using SETU email accounts, a credit-based parking system to ensure fair access, anonymous feedback mechanisms for academic evaluation, and marketplace for sharing educational materials.

Together, these components create a supportive ecosystem that promotes fairness and student success.

Overall, the system aims to accomplish a more connected and efficient university environment, where users can easily access essential services while contributing to a collaborative academic culture.

Features

SETU Student Support Services (S4):

- Login with SETU email to be able to use the features.
- The menu presents the 3 different sections, clicking on any brings you there.

Prioritized Parking Section:

- **Profile Dashboard:** Once passed the initial page, a dashboard will present itself. There will be a display of your default / initial credit count (which decreases corresponding to bookings placed). Every week the credit count will reset. Below and centered in the GUI, the user will see a “make a booking” option / bar.
- Booking Steps:
 - The next view will reveal a list of the SETU campus to choose (SETU CORK ROAD / COLLEGE ROAD)
 - The next view will be a calendar of the current week with toggle arrows implying the user can book as far as 7 days from now (in case the user is making a booking on Thursday for instance, otherwise they’d be limited to Friday).
 - In the day columns, there’ll be a status displaying whether there’s availability or not. Once a day is selected, the user will be placed on the final transactional view. Card details can be saved for user efficiency in future. After details are entered the user is prompted with at the end of the page, confirm booking.
 - After the booking process is completed, the user is sent back to the dashboard where a certain number of credits are indicated to have been used and an email, containing a QR code (or the like) is subsequently sent to the user to be scanned at the prioritised parking gate self-scanner.
- **Automatic Parking Spot Release:** If the user fails to show up on the day it booked, a penalty will be inflicted of a deduction of 4 points (initial weekly points are 6 points, two points per day, three days a week to book. Again this system ought to only cater to SETU users with specific reasons to get a parking spot, the credit system would be made redundant if a user could book a slot every day of the week which essentially a user shouldn’t have to). This means they will have lost the equivalent of two days booking credit fees. This is to prevent used spaces being on paper

vacant, when in actuality, other users' opportunities are being thwarted if the system reads no spaces when there practically is room.

- **Guest and Visitor Access:** Although SETU user authentication may suggest the service is exclusive, its not entirely based on strict conditions. Users on the system can book a slot on their behalf for a guest (guest speaker for career day for instance to represent a faculty or company etc). However, guest / visitor bookings are limited to consecutive, frequent use. The SETU user will be asked to provide the guest / visitors full name and email which then a personalised email will be sent for confirmation. This is to have a record of the person by which if this person is frequenting as a guest, the SETU user representing will receive a strict credit penalty.
- **Self – service portal:** In a menu bar above the dashboard there'll be an option to view 'my bookings'. Within this view a student will be posed with two options, cancel their booking or modify it to suit a different day for instance.

Rating Section:

- Rating system: Integers from 1 to 5 for each of the four categories.
 - For modules: enjoyment, material difficulty, exam/assignment difficulty, amount of work.
 - For lecturers: friendliness, fairness, difficulty, teaching ability.
 - All of them would have descriptions of what the rating means.
- In each comment, you can include up to 7 tags. Tags range from respectful, friendly, to unfair, lecture heavy.
- For each module/lecturer, there's an average score across each category, combining all categories, and a chart for score distribution.
- Rating can be made up to 4 times per user per thing, early/introduction (weeks 1 through 4), middle (weeks 5 through 8), end (weeks 9 through 12), and after (after week 12).
- You must include some additional information: time of rating (1 of the 4 available slots), international/domestic student, and some optional information, like grades of similar modules last semester.
- Ratings are anonymous to prevent bias towards students, but the data is kept behind the scenes for bonuses or regulatory purposes.

- Student Union would have a small department dedicated to regulating and publishing ratings. A rating isn't published until the SU has passed it through. This prevents destructive criticisms, trolls, and other unhelpful ratings.

Main Overview / Marketplace Section:

- This program will be made from the main marketplace where the students will be able to put up their resources for sale where students that might urgently need them could purchase the rights to them and download them.
- The UI for the program will be composed of 4 different views:
 - **Login/Logout/Register:** Here the user will be able to log in / log out or register to use the service; the user will need to provide their contact details.
 - The services will be encrypted to decrease the risks of data breaches.
 - The card details will be saved to improve the quality of life of users at checkouts.
 - **Profile dashboard:** The profile dashboard will provide the user with their:
 - Statistics
 - Their search history of resources where they can quickly return to the resources that they were looking for before,
 - Field where the user could change their details or communicate with the creator of the notes or other users to arrange study times, groups or just ask questions where they can get support.
 - There will also be their current balance. The user's balance will be used to purchase material in the marketplace.
- **Marketplace:** Marketplace is made up of just one main page with all their resources and filters for the resources on the right side, also a search bar on top to find exactly what someone is looking for.
- **Inventory:** This is the place where the purchased resources will appear for the users.
 - The resources will only be available in the app to remove the possibility of illegal distribution, so the resource will not be screen-recordable, screenshot-able or downloadable.

Product Vision

For SETU students and staff who want to help future students with ease of parking, the S4 (SETU Student Support Services) is an app that provides prioritized parking, lecturer and course rating, and information exchange. Unlike similar services, our product is simplistic to improve performance, straightforward, exclusive, user-friendly, and completely free.

Prioritized Parking Section:

1. *Flex Parking*

Positives:

- Strong self-service model. No manual administration needed.
- Suitable for organisations with large user bases (akin to SETU).
- Good operational efficiency through automation.

Negatives:

- Unaccustomed users might struggle without direct, familiar human support.
- Strong reliance on system stability.
- Difficult perhaps to customise for smaller organizations.

Features Ported:

- Self-service portal: This allows users to manage their own bookings and cancellations through automating routine tasks such as customer support.

2. *Ronspot*

Positives:

- Efficient real-time parking space management.
- Automated handling of unused / expired bookings.
- Strong access control via organisation-based authentication.
- Prevents misuse of reserved parking.

Negatives:

- Unforgiving to human fallibility (Strict time automated system).
- Legitimate delays resulting in lost bookings, potentially.
- Requires accurate monitoring to function properly.

Features Ported:

- **Auto-release:** It can automatically free up parking spaces if a user exceeds the time window it can arrive at their booked slot. The essential and fundamental direction of my app is that everyone, always (within SETU opening hours), has an opportunity to have a slot and that no fellow user can impede another user's experience.
- **User Authentication:** The admin can whitelist specific email domains. In this instance, '@setu / @wit.ie'. Therefore, the app is of exclusive service to affiliates of SETU.

3. Dibsido

Positives:

- Prevents long-term monopolisation by individual users.
- Encourages mindful booking behaviour.
- Works well in environments carrying high demand.

Negatives:

- May feel restrictive to frequenting users.
- Less suitable for users with consistent daily needs.
- Credit system adds to system complexity, intimidating to first-time users.

Features Ported:

- **Credit System:** Encourages the equitable principle of the app by establishing a system where at beginning of each month, a user has a default, starting amount of credits which are automatically deducted upon booking a slot. This goes against the notion that a user can have their name figurately on a parking slot each day of the working week, simply because they can.

4. Flexopus

Positives:

- Structured guest and visitor access.

- Upholds accountability through user-linked bookings, prevents exploitation.
- Balances internal and external access.
- Suitable for academic and corporate environments alike.

Negatives:

- Booking process is more time-consuming.
- Relies on accurate information entry.
- Guest limitations might perhaps frustrate repeated visitors.

Features Ported:

- **Guest Access:** A guest system feature where an affiliate of the college can book a slot for a guest (a guest speaker to represent a company, a faculty etc) who often must arrive at the college. However, guest bookings are limited against daily / consecutive use.

Rating Section:

1. *RateMyProfessor*

Positives:

- Simple, nice-looking UI. Understandable and straightforward.
- Each comment can add adjective tags, and the most popular tags are displayed. Good system.
- Quality rating are integers from 1-5, with an added difficulty rating also from 1-5.
- Also rates schools/universities. Curiously, I couldn't find SETU.
- Indicates course/module for each comment, in addition to some other information like online class, textbook, would take again, decently comprehensive.
- Ratings are all anonymous.

Negatives:

- The difficulty system can be conflated with course difficulty.
- Lacks a course difficulty section.

- Little to no moderation, so some comments aren't very helpful.
- No sorting system for comments, it's eternally sorted by most recent.
- Doesn't say which year each commenter attended.
- Character cap for each comment is too low.
- No verification for who posts. Could be subject to review bomb or review boost. You don't even need to login to rate.

Features Ported:

- Bits of the UI. Rating distribution chart, page organization are great.
- The tagging system.
- Anonymous rating, but my version still has the user data stored in the backend.

Note: Similar idea, but mine's more focused onto SETU. RMP seems to be giving users information briefly, while mine goes deeper.

2. *StudentCrowd*

Positives:

- Very comprehensive. Has ratings for universities, courses, accommodations, and locations.
- Detailed for each course what you need to achieve to get in and what to expect.
- Articles for people who are considering it.
- Rating system has identification to a degree. It asks if you're a domestic or international student, or if you're a prospective student, parent.
- It probably verifies you are from that university before submitting. I didn't try, but it asked for my university email.
- Has 5 different criteria when rating, each criteria has 5 different levels and a small description for each. Very nice looking.
- Anonymous ratings.

Negatives:

- Commenting and rating is barebones. It's functional and identifies the correct individual, but no interesting systems.
- UI to see the comments aren't very good. The main thing you get is an average score, which isn't much to go off on.
- Also, no sorting, but no filtering as well.
- No year of attendance.
- SETU is on here, but there's not enough reviews to get an average score.
- A completely different vibe compared to my system. This seems to try and inform purely future students who aren't in this university/course.
- Mid looking website.

Features Ported:

- Criteria with levels and descriptions.
- Domestic vs international student verification.

Note: Even broader compared to RMP, so even further from my idea. My concept seems to fulfil a niche role, which was by design, since it's very targeted.

Marketplace Section:

1. WhatsApp groups

Positives:

- Direct communication between members
- Highly customizable by the admin of the group when it comes to members
- Very secure
- High availability
- Large customer base
- Free

Negatives:

- No quality control
- Information lost in the chat
- No admin organisation capabilities
- No built-in rating system

Features Ported:

- High availability
- Security
- Direct communication

Note: WhatsApp was created with communication in mind, not a course material marketplace.

2. Course Hero

Positives:

- Large global database of materials
- Multiple filters for material
- Good review and rating system
- Quality control
- Simple UI

Negatives:

- Expensive subscription fees
- Not specific to just one institution
- No guarantee of materials for a certain institution
- Might violate academic integrity

Features Ported:

- All the positives will be implemented in our app.

Notes: They will be implemented without the emphasis on the large global database of resources. The students and the university will be prioritised over the platform.

Personas

What is a Persona?

In Software Engineering, a semi-fictional, data-driven profile representing a specific segment of a target user base.

Rating Section:

1.

Name: Bao Gia Tran Pham

Age: 18

Role / Occupation:

First year student at SETU Cork Road campus, studying Computer Science.

Background:

International student from Vietnam. Finds the Irish academic culture very different from what he's used to. Is adjusting by asking seniors and just improvising. Wants to have a centralized way of getting such information.

Goals & Motivations:

Wants to have detailed, constructive reviews that are useful to students from non-EU backgrounds. And hopefully leave some in the future when he's fully adjusted.

Quote:

"How do they grade your assignments? How strict are they? How much work is there? How fun is the lecture? It's the same questions to every lecturer."



2.

Name: Emma Clarke

Age: 21

Role / Occupation:

Second-year Arts student at SETU Cork Road campus.

Background:

Came from Kilkenny, has no idea what optional module to pick up for the second semester. Senior and lecturer advice clashes and don't converge on anything.

Goals & Motivations:

Wants an honest, structured way to scope out modules before picking. Trusts peer reviews over official course descriptions.

Quote:

“The module descriptor says, ‘challenging and rewarding’, like every other one. But I want to know if the exam is doable and if the lecturer is friendly or not.”



3.

Name: Dr. Peter Murphy

Age: 45

Role / Occupation:

Senior lecturer in Business, 14 years at SETU.



Background:

Takes his teaching seriously and invests a lot of time in updating his material each semester. Informal feedback is very inconsistent, while end-of-year module surveys have low response rates and still feel too polite.

Goals & Motivations:

Would like a transparent, ongoing feedback channel. Interested in seeing tagged feedback along with numeric scores.

Quote:

“Look, it is nice to hear ‘this was a good class’, but it’s just too vague! What about it is good? Nothing bad?”

4.

Name: Ryan Cooper

Age: 22

Role / Occupation:

Final year student at SETU Cork Road campus, studying Business.
a student ambassador.

Background:

Regularly he hears questions from prospective students and parents on open days about what the course is really like and is frustrated that he can only give his own subjective opinion, since it changes year by year.

Goals & Motivations:

Wants a trustworthy reference that can be shared with concerned outsiders that reflects the current sentiment. It should be deep enough to be hard to make results unfair and not a marketing copy.

Quote:

“If we have such a rating system, I think people would be happier to see a honest, anonymized rating system. At least I will be!”



Prioritized Parking Section:

5.

Name: Ciara Murphy

Age: 21

Role / Occupation:

Final year Computing student at SETU Cork Road campus.



Background:

She lives 18 km outside of Waterford in a rural area with limited public transport options. Drives a typical young driver second-hand car. She Has important final exams and final year project due soon and mandatory lab sessions starting at 9am. Ultimately, frequently arrives stressed after circling the car park for 20-30 minutes.

Goals & Motivations:

Guarantee a parking spot on critical days (exam weeks, presentations etc). Avoid the risk of being late or missing crucial tasks entirely. She hopes to plan ahead, during mid-week for the following week.

Quote:

“I can’t risk not having a space on assessment days, those 20 minutes driving around could cost me my module.”

6.

Name: Dr. Gerald Doyle

Age: 38

Role / Occupation:

Course leader, lecturer in Computing & guest coordinator in the Science and Computing.



Background:

Commutes daily from Carrick-on-Suir, Tipperary. Often hosts guest speakers, industry visitors etc. His own opportunity of parking can be limited, and he often needs to secure spots for visitors to make a good impression of the college's management and to avoid complicating the guest's arrival.

Goals & Motivations:

Ensures reliable parking for himself and his guests on crucial days (e.g., career fair speakers, company reps etc). Wants efficiency and professionalism.

Quote:

"Having confirmed a spot for a visiting CEO or senior employee makes all the difference and takes a big weight off my job. No one wants to witness their guest which took time out for them aimlessly circulating the parking space for a 9am talk."

7.

Name: Sarah Brennan

Age: 34

Role / Occupation:

Part – time mature student in Health Sciences. Works 20 hours per week as a healthcare assistant; single parent with two children attending school.

Background:

Drives from Ferrybank. Has drop off and pick up responsibilities, so timing for Sarah is critical. Only attends campus 3 days a week for lectures and clinical labs. Can not afford to be late or miss classes on account of family and work commitments.

Goals & Motivations:

Secure parking on her specific campus days to avoid being late for work and family demands. Sarah needs predictability to balance responsibilities.

Quote:

“Between lectures, work and school runs, I just need things to go my way, knowing I have a parking spot means I won’t have to panic on a daily basis.”



Marketplace Section:

8.

Name: Isak Wahlqvist

Age: 22

Role / Occupation:

International student that just graduated in electrical engineering at SETU.

Background:

Just finished his education, because he is in the middle between transitioning from studies to work, he has a lot to do and needs money.

Goals & Motivations:

Isak has a lot of notes from his last 4 years of studying, since he is in need for money he is looking for a way to monetize his old notes and material.

Quote:

“I just want to close the college chapter and maybe earn some money.”



9.

Name: Dr. Elena Ekmann

Age: 41

Role / Occupation:

Lecturer in Macroeconomics at SETU, 3 years at SETU

Background:

Internationally recognized researcher and PhD in macroeconomics holder from Austria. She has worked for Austrian mint and worked closely with ECON (European Parliament's Committee on Economic and Monetary Affairs)

Goals & Motivations:

She is going to voluntarily moderate the system to remove any kind of plagiarism as it might be considered “receiving or possession of stolen goods.” which is a criminal offense, and she takes it very seriously.

Quote:

“Honest economics are an integral part of student life”



10.

Name: Serj Tankian

Age: 58

Role / Occupation:

A Software Manager at SETU Waterford.



Background:

He is a manager who also has qualifications in software systems engineering. This makes him perfect for the position of managing software systems at SETU. He previously studied at UCC and got his master's degree in management and marketing at the same institution.

Goals & Motivations:

As a software manager at SETU, he is willing to help set up and maintain the platform if it is not globally monetized and improves the feeling of unity between students where everyone can contribute and make little side money.

Quote:

“Well maintained and managed software is a pillar of education institution.”

Scenarios

What is a Scenario?

a specific, realistic situation or sequence of user actions/system behaviours used to evaluate, test, or define software functionality.

Rating Section:

Scenario 1:

Bao Gia Tran Pham is a first-year student halfway through the first semester. The course leader informed him that everyone should start thinking about what stream they want to pick for the second semester and provide short informational videos from each stream. Choosing a stream is integral to the course and will be permanent from year 2.

Even though he watched each video multiple times, he still can't decide between cybersecurity and cloud. Job prospects sound good for both, but Bao was unsure on how hard the course is. That's when he remembered S4.

After emailing the course leader for the name of the modules and lecturers, Bao spent an hour on S4's LMR section. He then decided cloud seemed like a better stream from the countless people praising its stream-specific-module. He submitted the form to the course leader the next day and has never regretted his decision.

Scenario 2:

Dr. Peter Murphy is a senior lecturer in Business and has recently shifted his curriculum massively to integrate more recent developments into the module. Though his fellow lecturers reassure him it's good, he doesn't know if they're just being polite, or if it's good.

At the end of each week, he goes on S4's LMR section to see what students are talking about. There's always a big influx of reviews at the end of each of the 3 submission periods, which gives Peter a nuanced view of his curriculum. He sees that even the enthusiastic students dislike the number of assignments given out, and the weaker students struggle with his speed. Slightly disappointed but glad he checked it out, Peter reviews and shifts the curriculum. Students' reception improved noticeably in the following year.

Scenario 3:

Ryan Cooper is a final year student studying Business while being a student ambassador. A new batch of prospective students are coming next week, and Ryan feels conflicted about his course experience. Though he enjoyed it, his friends were struggling, and quite a few dropped out.

After he asked for advice, his lecturer, Dr. Peter, told him to look at the new service SETU started 2 years ago. Previously, he thought it was something unrelated or just for tech students. But the LMR section had exactly what he needed. He saw that most of the feedback for his year was like his friends, and he was the odd one out.

The following week, Ryan happily shared the service's webpage with the nervous prospective Business students. He also made sure to thank Dr. Peter in person.

Marketplace Section:

Scenario 4:

Isak Wahlqvist just graduated and got his bachelor's degree in electrical engineering. With this new achievement, he is ready to find employment and start his career within the field that he studied for the last 4 years. Unfortunately, the job market is very rough right now and it looks like he might struggle a lot to find a job for quite a while now.

After carefully considering his options as of now he decides to keep looking for work while making any money he can on the side. Since he has been attending large majority of his classes and has had really good results in his modules, he is looking for ways to monetize these past efforts of his right now by either tutoring or selling his materials.

Isak found the S4's information exchange section to be very useful. He took his notes and put them up in the marketplace. This helped his finances a lot since all notes were readable and reliable for other students. The notes were also all connected to each other, making somewhat of an ecosystem which students utilized to better understand the topic and gave those notes high ratings. This helped both the current STEM students and Isak achieve their goals.

Scenario 5:

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Mace Windu is currently pursuing his bachelor's degree in law. Due to his high engagement in extracurricular activities and the high difficulty of the course he needs to balance all those activities while having an income and not sacrificing his social life. This is a large struggle for him since he has a lot to do.

During his education he came across S4's marketplace platform, which at first he didn't care at all about but with time and after giving it a chance he realized he could save a lot of time by putting up his notes and getting notes from there. Since he can buy notes from others, he has more time for other activities and with his original notes being monetizable there he breaks even and doesn't need to spend as much time in classes which helps his cause.

While using the app, he finds it useful to filter out material and see it with reference to module and lecturer. This helps him understand and find patterns in what the lecturer finds important and improves his test scores a lot.

Scenario 6:

Serj Tankian is a software manager at SETU. He wants the school to perform as good as possible to drive in new revenue while helping young adults to get the education they dream of. With this new platform, students' time would be optimized, leaving them more free time for doing whatever they want while not sacrificing their academic results. Serj really puts a lot of trust into this new platform and manages it to improve the enjoyability of studying and QS rankings at SETU.

He is very fond of the cooperation that students can achieve in this platform while learning about economics and how the market works hands on. He is very willing to manage this piece of software.

Prioritized Parking Section:

Scenario 7:

Ciara Murphy, a Final-year Computing Student, lives 18km away with limited transport. Ciara has a 9:00 AM mandatory lab assessment and cannot risk being late.

During the middle of the week, Ciara logs into S4's prioritized parking section to plan for the following week. She uses her credits to reserve a space for the big assessment. As a result, on the morning of the lab, she drives directly without any complications to her reserved spot at the Cork Road campus, arriving calm and on time as opposed to "stressed off my head on a frequent basis".

Scenario 8:

Dr. Gerard Doyle, course leader and guest co-ordinator. Dr. Doyle is hosting a visiting CEO for a 9:00 talk. He wants to steer clear of the weight on his back of the case of his guest likely to fall into aimlessly circulating the campus car park, which would reflect poorly on the colleges management.

After Gerard registered the CEO on S4's prioritized parking section, the CEO arrives and parks almost instantly. This efficiency ensures to deliver a professional first impression and allows Dr. Doyle to focus on the career event more strongly without lingering in the back of his mind, parking logistics.

Scenario 9:

Sarah Brennan, part-time mature student + single parent. Sarah has strict time management, surrounding her maternal responsibilities picking up or dropping off her children. She only attends college three days a week and has no issue at all balancing the two responsibilities.

Sarah uses S4's prioritized parking system to guarantee a spot on her three specific campus days. It's imperative she balances her family and work commitments. Knowing she has a guaranteed spot, Sarah doesn't have to panic and the transition between the aforementioned sides of her life can run smoothly.

User Stories

What is a User Story?

a short, informal description of a software feature written from an end-user's perspective to articulate what they need and why.

Rating Section:

Bao Tran Pham Gia:

- As a freshman, Bao wants to get an objective measurement on quality of lecturers and modules.
- As someone picks their stream, Bao wants to get a feel of each stream from unbiased reviews.

Dr. Peter Murphy:

- As a dedicated lecturer, he wants to know what students thought of his lectures.

Ryan Cooper:

- As a student ambassador, he wants to let prospective students know how good the modules are.
- As a final year student, he wants to know opinions about the course from people outside his friend circle.

Marketplace Section:

Isak:

- As a past student I need to be able to find more use in all the notes I have after graduating.
- As a graduate from SETU Isak might be able to share some of his experience within the platform to improve the quality of studies for the current students.

Mace Windu:

- As a current student I really need more time so using the portal to purchase material and save me some time is great.
- As a current student I also need to be able to distinguish between what lecturers might want from me as a student, looking at past notes and finding patterns helps with that a lot.

Serj Tankian:

- As a software manager for SETU it's important for him to make new ways for students to help each other out digitally which is promoted with this platform.

Prioritized Parking Section:

Ciara:

- As a student with limited means of transportation, such as traffic, and a mandatory assessment, I want to reserve a parking space in advance so that I can avoid delays and arrive on time for my exams.

Gerard:

- As a course leader hosting an important guest, I need to pre-allocate a parking space for my visitor so that the college maintains a professional reputation and the event starts on schedule.

Sarah:

- As a student and a mother with strict family and time commitments, I want to guarantee a parking spot for my specific campus days so that I can effectively balance my parental responsibilities with my education.

Prototype

What is a prototype?

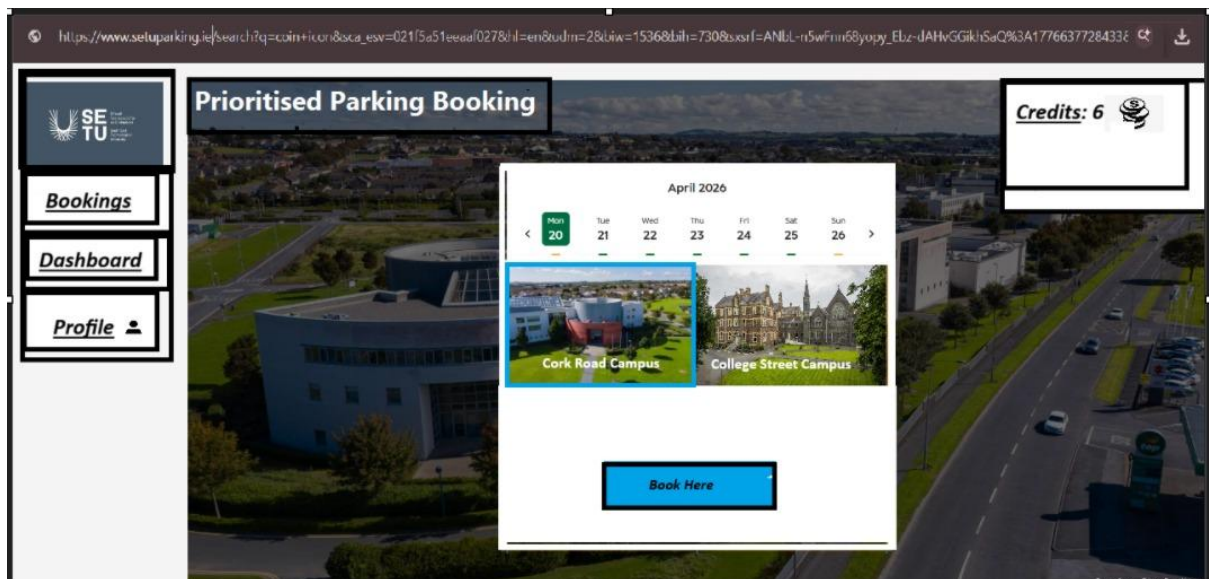
an early, preliminary version of a software product used to test concepts, visualize designs, and gather user feedback before full-scale development.

What tool did you use?

Scene builder (+ Microsoft paint).

UI:

Prioritised Parking:



Marketplace Section:

The screenshot displays the 'School Resources Marketplace' interface. At the top, there is a search bar with the placeholder text 'Search notes, past papers, books...' and a 'Sort by:' dropdown menu. On the left side, there are two filter sections: 'Categories' and 'Subjects'. The 'Categories' section has three checkboxes: 'Notes & Summaries', 'Past Papers', and 'Study Guides'. The 'Subjects' section has six checkboxes: 'SE600' (checked), 'SE703' (checked), 'SE500', 'SE444', 'SE603', and 'SE301'. Below these is a 'Price' slider ranging from 0 to 100. The main content area on the right displays three resource cards. The first card is titled 'Mathematical Methods' for 'SE600 - Semester 3', describing 'Basic linear algebra with worked examples', priced at '€3.99' with a 'Get' button. The second card is titled 'Computer Systems I' for 'SE600 - Semester 1', describing 'Linux lab - all commands in course used', priced at 'FREE' with a 'Get' button. The third card is titled 'Construction Technology I' for 'SE703 - Semester 1', describing 'Worked examples covering concrete build-ups', priced at '€4.99' with a 'Get' button.

School Resources Marketplace

Search notes, past papers, books... Sort by: ▾

Categories

- ☐ Notes & Summaries
- ☐ Past Papers
- ☐ Study Guides

Subjects

- ☒ SE600
- ☒ SE703
- ☐ SE500
- ☐ SE444
- ☐ SE603
- ☐ SE301

Price

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Rating Section:



ModuleLecturer

3.78^{/5}

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DATA STRUCTURES AND ALGORITHMS 2
Lecturer: Peter Carew

Rate

Points Explanation

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Year: 2026 - 2027 Period: Early International: Yes ...

E: 5 Easy SAQs, no homework. Assignments are a breeze. MCQ is difficult though, similar to DSA 1, material is understandable.
M: 2
A: 4 #EasyPass, #HardMCQ, #Simple
W: 1

April 18th, 2026

Enjoyment

3.78

Material Difficulty

1.23

Assessment Difficulty

3.89

Work Amount

2.13

Top 3 Tags

Easy Pass 92

Hard MCQ 88

Complex 86

Main Page:

Welcome to SETU Student Support Services! Please select an option...



Reserve your parking slot!

Prioritized Parking



Rate lecturers and modules!

LMR



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Marketplace

Tests

Rating Section:

Main:

```
package main;

//TIP To <b>Run</b> code, press <shortcut actionId="Run"/> or
// click the <icon src="AllIcons.Actions.Execute"/> icon in the gutter.

public class Main {

    public static void main(String[] args) {

        //TIP Press <shortcut actionId="ShowIntentionActions"/> with your caret at the highlighted text
        // to see how IntelliJ IDEA suggests fixing it.

        System.out.printf("Hello and welcome!");

        for (int i = 1; i <= 5; i++) {

            //TIP Press <shortcut actionId="Debug"/> to start debugging your code. We have set one <icon
            src="AllIcons.Debugger.Db_set_breakpoint"/> breakpoint
            // for you, but you can always add more by pressing <shortcut actionId="ToggleLineBreakpoint"/>.

            System.out.println("i = " + i);

        }

    }

}
```

Back – End:

```
import java.util.ArrayList;

import java.util.Arrays;

import java.util.HashMap;


import static java.util.Arrays.sort;


public class SuperComplexBackEnd {

    private ArrayList <String> emails=new ArrayList();

    private HashMap <String, HashMap <String, Integer>> tags=new HashMap();


    public void addAllEmails(String[] emails){ //Used for set up tests

        this.emails.addAll(Arrays.asList(emails));

    }


    public String addEmail(String email) {

        if (emailCheck(email).equals("OK")) {
```

```
        emails.add(email);

        return "OK";
    }

    return emailCheck(email);
}

public boolean containsEmail(String email) {
    return emails.contains(email);
}

public void addTag(String name, String tag) {
    if (!tags.containsKey(name))
        tags.put(name, new HashMap<>());

    if (tags.get(name).containsKey(tag))
        tags.get(name).put(tag, tags.get(name).get(tag) + 1);
    else
        tags.get(name).put(tag, 0);
}

public void setTagCount(String name, String tag, int count) {
    if (!tags.containsKey(name))
        tags.put(name, new HashMap<>());

    tags.get(name).put(tag, count);
}

public ArrayList<String> getTopThreeTags(String name) {
```

```

if (!tags.containsKey(name))

    return null;

int[] tmpArr=new int[tags.get(name).size()];

int index=0;

for (Integer i : tags.get(name).values()) {

    tmpArr[index++]=i;

}

sort(tmpArr);

ArrayList<String> topThreeTags=new ArrayList();

for (int i=index-1; i>=Math.max(index-3, 0); i--) {

    for (String s : tags.get(name).keySet()) {

        if (tags.get(name).get(s) == tmpArr[i] && !topThreeTags.contains(s)) {

            topThreeTags.add(s);

            break;

        }

    }

}

if (topThreeTags.isEmpty())

    return null;

return topThreeTags;

}

public String emailCheck(String email) {

    if (email.endsWith("@setu.ie"))

    {

        if (!containsEmail(email))

```

```
        return "OK";

        return "Email already exists";

    }

    return "Email does not belong to SETU";

}

public void clearEmails() {

    emails.clear();

}

public void clearTags() {

    tags.clear();

}

}
```

Tests:

```
import org.junit.jupiter.api.AfterEach;

import org.junit.jupiter.api.BeforeEach;

import org.junit.jupiter.api.Test;

import java.util.ArrayList;

import static org.junit.jupiter.api.Assertions.*;

class baoTests {
```

```

SuperComplexBackEnd backEnd=new SuperComplexBackEnd();

@BeforeEach

void setUp() {

    backEnd.addAllEmails(new String[] {"20108791@setu.ie", "lecturer1@setu.ie", "student1@setu.ie",
        "lecturer2@setu.ie", "lecturer3@setu.ie", "student2@setu.ie"});

    backEnd.setTagCount("Trish", "Friendly", 27);

    backEnd.setTagCount("Trish", "Lenient Scoring", 25);

    backEnd.setTagCount("Trish", "Fair", 22);

    backEnd.setTagCount("Trish", "Nice", 21);

    backEnd.setTagCount("Trish", "Harsh Scoring", 1);

    backEnd.setTagCount("Peter", "GOAT", 15);

    backEnd.setTagCount("Jerry", "Hard MCQ", 100000);

    backEnd.setTagCount("Jerry", "Harsh Scoring", 100000);

}

@AfterEach

void tearDown() {

    backEnd.clearEmails();

    backEnd.clearTags();

}

@Test

void emailTest() {

    //Adding valid email

    assertEquals("OK", backEnd.addEmail("20113304@setu.ie"));

```

```

assertTrue(backEnd.containsEmail("20113304@setu.ie"));

//Adding invalid email - outside email

assertEquals("Email does not belong to SETU", backEnd.addEmail("randomGoofball@gmail.com"));

assertFalse(backEnd.containsEmail("randomGoofball@gmail.com"));

//Adding invalid email - duplicate email

assertEquals("Email already exists", backEnd.addEmail("20113304@setu.ie"));
}

@Test
void tagTest()
{
    //Test top 3 with >3 entries

    ArrayList<String> top3Trish=backEnd.getTopThreeTags("Trish");

    assertTrue(top3Trish.contains("Fair"));

    assertTrue(top3Trish.contains("Friendly"));

    assertTrue(top3Trish.contains("Lenient Scoring"));

    System.out.println("Trish\n");

    for (String tag : top3Trish)

        System.out.println(tag);

    //Test adding more tag counts

    backEnd.addTag("Trish", "Nice");

    backEnd.addTag("Trish", "Nice");

    top3Trish=backEnd.getTopThreeTags("Trish");

```

```

assertTrue(top3Trish.contains("Nice"));

assertFalse(top3Trish.contains("Fair"));

//Test top 3 with <3 entries

ArrayList<String> top3Peter=backEnd.getTopThreeTags("Peter");

System.out.println("\n\nPeter\n");

for (String tag : top3Peter)

    System.out.println(tag);

assertTrue(top3Peter.contains("GOAT"));

assertEquals(1, top3Peter.size());

//Test equality handling

ArrayList<String> top3Jerry=backEnd.getTopThreeTags("Jerry");

System.out.println("\n\nJerry\n");

for (String s : top3Jerry) System.out.println(s);

assertTrue(top3Jerry.contains("Hard MCQ"));

assertTrue(top3Jerry.contains("Harsh Scoring"));

assertEquals(2, top3Jerry.size());

//Test non-existent

assertNull(backEnd.getTopThreeTags("Joey"));

}

}

```

Prioritised Parking:

Main:

```
//TIP To <b>Run</b> code, press <shortcut actionId="Run"/> or  
// click the <icon src="AllIcons.Actions.Execute"/> icon in the gutter.  
  
public class Main {  
  
    public static void main(String[] args) {  
  
        //TIP Press <shortcut actionId="ShowIntentionActions"/> with your caret at the highlighted text  
        // to see how IntelliJ IDEA suggests fixing it.  
  
        System.out.printf("Hello and welcome!");  
  
        for (int i = 1; i <= 5; i++) {  
  
            //TIP Press <shortcut actionId="Debug"/> to start debugging your code. We have set one <icon  
src="AllIcons.Debugger.Db_set_breakpoint"/> breakpoint  
  
            // for you, but you can always add more by pressing <shortcut actionId="ToggleLineBreakpoint"/>.  
  
            System.out.println("i = " + i);  
  
        }  
  
    }  
}
```

Back-End

```
import java.util.ArrayList;

import java.util.HashMap;

public class BackEnd {

    private ArrayList <ParkingSlot> bookings=new ArrayList();

    private HashMap <String, ArrayList <Integer>> parkingSlots=new HashMap <>();

    private HashMap <String, Integer> userCredit=new HashMap();

    public void setCredit(String name, int num)

    {

        userCredit.put(name, num);

    }

    private boolean checkCredit(String name)

    {

        if (!userCredit.containsKey(name))

            setCredit(name, 10); //Initial Credit

        return userCredit.get(name)>=3;

    }

    public boolean addBooking(String name, ParkingSlot booking) {

        if (!parkingSlots.containsKey(name))

            parkingSlots.put(name, new ArrayList <>());

        if (!checkExist(booking) && checkCredit(name)) {
```

```
        setCredit(name, userCredit.get(name)-3);

        bookings.add(booking);

        parkingSlots.get(name).add(booking.id);

        return true;
    }

    return false;
}

public ParkingSlot getBookingById(int id) {

    for (ParkingSlot booking : bookings) {

        if (booking.id == id)

            return booking;

    }

    return null;
}

private void removeBooking(ParkingSlot booking) {

    for (ParkingSlot book : bookings) {

        if (book.id==booking.id && book.day==booking.day && book.month==booking.month &&
book.year==booking.year) {

            bookings.remove(book);

            return;

        }

    }

}
```

```
public boolean modifyBooking(String name, ParkingSlot oldBooking, ParkingSlot newBooking) {  
  
    if (!parkingSlots.containsKey(name) || !parkingSlots.get(name).contains(oldBooking.id))  
  
        return false;  
  
    if (checkExist(oldBooking) && !checkExist(newBooking)) {  
  
        removeBooking(oldBooking);  
  
        bookings.add(newBooking);  
  
        parkingSlots.get(name).add(newBooking.id);  
  
        for (Integer i : parkingSlots.get(name)) {  
  
            if (i == oldBooking.id) {  
  
                parkingSlots.get(name).remove(i);  
  
                break;  
  
            }  
  
        }  
  
        return true;  
  
    }  
  
    return false;  
  
}  
  
public boolean checkExist(ParkingSlot booking) {  
  
    for (ParkingSlot slot : bookings) {  
  
        if (slot.day == booking.day && slot.month == booking.month && slot.year == booking.year)  
  
            return true;  
  
    }  
  
    return false;  
  
}
```

```
public void clearBookings() {  
  
    bookings.clear();  
  
    parkingSlots.clear();  
  
}  
  
public void clearCredits() {  
  
    userCredit.clear();  
  
}
```

Parking Slot

```
public class ParkingSlot {  
  
    public int id, day, month, year;  
  
    public ParkingSlot(int id, int day, int month, int year) {  
  
        this.id = id;  
  
        this.day = day;  
  
        this.month = month;  
  
        this.year = year;  
  
    }  
  
}
```

Tests

```
import org.junit.jupiter.api.AfterEach;
```

```

import org.junit.jupiter.api.BeforeEach;

import org.junit.jupiter.api.Test;


import java.util.ArrayList;


import static org.junit.jupiter.api.Assertions.*;

public class JoeyTests {

    BackEnd backEnd=new BackEnd();


    @BeforeEach

    void setUp() {

        backEnd.addBooking("Joey", new ParkingSlot(1, 31, 9, 2026));

        backEnd.addBooking("Bao", new ParkingSlot(2, 12, 2, 2027));

        backEnd.addBooking("Bao", new ParkingSlot(3, 15, 6, 2027)); //Bao has 4 credits

        backEnd.addBooking("Joey", new ParkingSlot(4, 23, 4, 2027));

        backEnd.addBooking("Joey", new ParkingSlot(5, 30, 1, 2027)); //Joey has 1 credit

        backEnd.addBooking("Gustaw", new ParkingSlot(6, 6, 2, 2027)); //Gustaw has 7 credits

    }

    @AfterEach

    void tearDown() {

        backEnd.clearBookings();

        backEnd.clearCredits();

    }


    @Test

    void makeBooking() {

```

SEP ASSIGNMENT 2026

```
//Check valid booking

assertTrue(backEnd.addBooking("Bao", new ParkingSlot(7, 31, 8, 2027))); //Bao has 1 credit

assertTrue(backEnd.checkExist(new ParkingSlot(7, 31, 8, 2027)));


//Check invalid booking - duplicate

assertFalse(backEnd.addBooking("Gustaw", new ParkingSlot(8, 31, 8, 2027)));

assertNull(backEnd.getBookingById(8));


//Check invalid booking - not enough credits

assertFalse(backEnd.addBooking("Bao", new ParkingSlot(9, 31, 8, 2028)));

assertNull(backEnd.getBookingById(9));
}


@Test
void checkModifyBooking() {

    //Valid rebook

    assertTrue(backEnd.checkExist(new ParkingSlot(5, 30, 1, 2027)));

    assertTrue(backEnd.modifyBooking("Joey", new ParkingSlot(5, 30, 1, 2027), new ParkingSlot(5, 5, 2,
2028)));

    assertFalse(backEnd.checkExist(new ParkingSlot(5, 30, 1, 2027)));

    assertTrue(backEnd.checkExist(new ParkingSlot(5, 5, 2, 2028)));


    //Invalid rebook

    assertTrue(backEnd.checkExist(new ParkingSlot(1, 31, 9, 2026)));

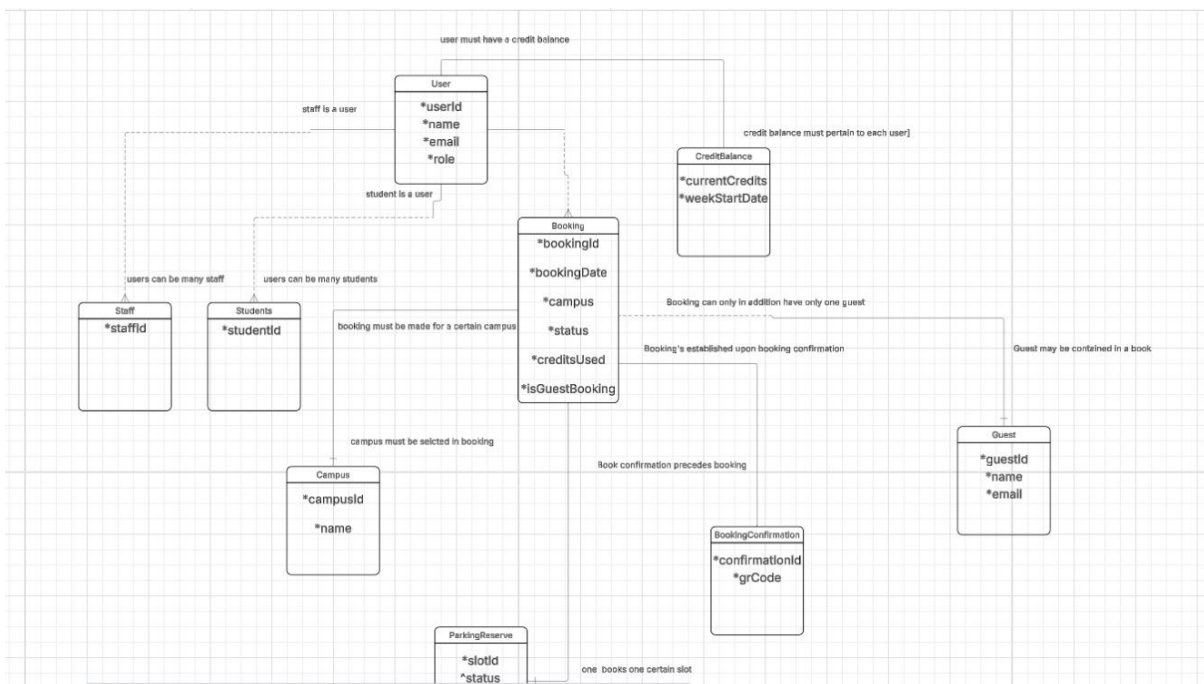
    assertFalse(backEnd.modifyBooking("Joey", new ParkingSlot(1, 31, 9, 2026), new ParkingSlot(1, 12, 2,
2027)));

    assertTrue(backEnd.checkExist(new ParkingSlot(1, 31, 9, 2026))); //Should still be there
```

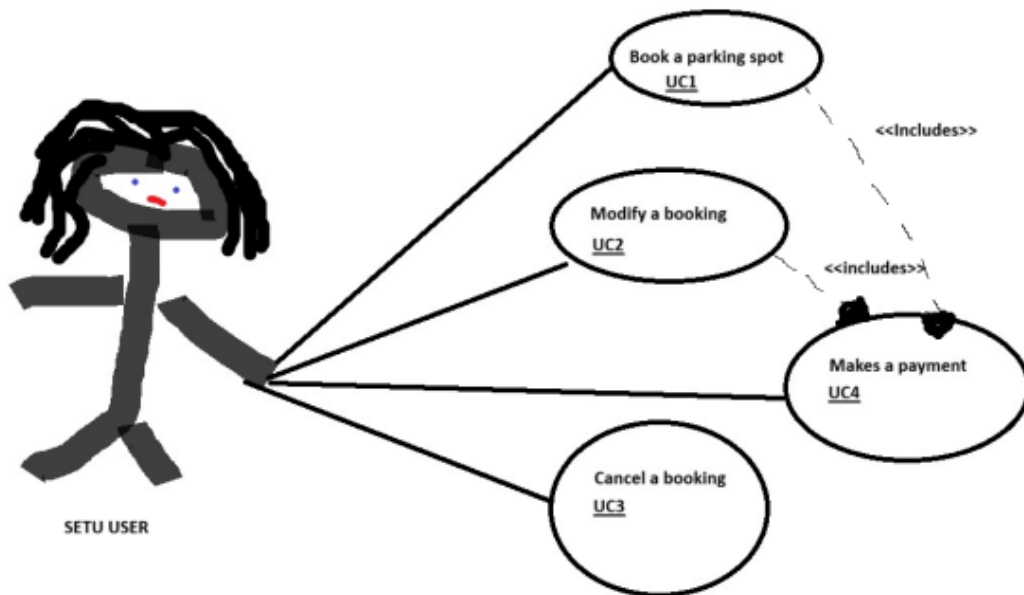
```
assertFalse(backEnd.checkExist(new ParkingSlot(1, 31, 9, 2027)));
```

Diagrams

Class Diagram:



Use Case Diagram:



UC1: BOOK A PARKING SLOT

Use case: Book a prioritised parking slot in advance

Primary actor: SETU User (Student or Staff)

Preconditions:

- User has navigated to the SETU parking booking website.
- User has registered for an account and signed in with a valid SETU email (@setu.ie or @wit.ie)

Main success scenario

1. System displays the dashboard showing current weekly credit balance (default = 6 credits)
2. User selects “Make a Booking”
3. System displays list of campuses (Cork Road / College Road)
4. User selects a campus
5. System displays calendar for the current week with toggle arrows (maximum 7 days ahead)

6. User selects a day and sees availability status
7. System displays available time slots / parking spots for that day
8. User selects a preferred parking slot
9. System checks credit balance and displays summary (credits that will be deducted)
10. User confirms selection and proceeds
11. System processes the booking and deducts the required credits
12. System generates a QR code
13. System sends confirmation email containing the QR code
14. System returns user to the dashboard with updated credit balance

Alternative success scenarios

User books for a Guest

- At step 8, user chooses “Book for Guest” option
- System asks for guest’s full name and email
- System sends separate confirmation email to the guest
- Booking proceeds as normal (credits deducted from the SETU User’s account)

Automatic seat/spot assignment

- At step 7, user selects “Assign any available spot” instead of choosing manually

User changes mind at any time

- a. User presses Cancel
- b. System prompts for confirmation
- c. User confirms
- d. System returns to step 2 (Make a Booking screen)

Failure scenarios

Insufficient credits At step 9:

- a. System displays “Insufficient credits – you need X more credits”
- b. System offers option to wait for weekly reset or cancel

- c. User is returned to dashboard

No availability

At step 7:

- a. System displays “No spots available on selected day”
- b. User is invited to choose a different day or campus

Invalid campus/date

At step 6:

- a. System displays error if date is in the past or more than 7 days ahead
- b. User is prompted to select a valid date

UC2: MODIFY A BOOKING

Use case: Modify an existing parking booking

Primary actor: SETU User

Preconditions:

- User has already made a booking
- User has a valid booking reference (or can view it in “My Bookings”)
- User has signed in

Main success scenario

1. User navigates to “My Bookings” from the menu
2. System displays list of current and upcoming bookings
3. User selects the booking they wish to modify
4. System displays current booking details
5. User selects “Modify Booking”
6. System displays calendar (same 7-day limit) and available spots
7. User selects new date and/or parking slot

8. System calculates any credit difference or penalty
9. System displays summary of changes and any extra credits required
10. User confirms changes
11. System processes the modification and updates the booking
12. System sends confirmation email with updated QR code (if changed)
13. System returns user to “My Bookings”

Alternative success scenarios

No credit change required

- At step 9, system shows “No additional credits required”
- User confirms and skips payment

User changes mind

- User presses Cancel at any point
- System returns to step 2

Failure scenarios

New slot no longer available

At step 7:

- a. System shows error “Slot is no longer available”
- b. User is asked to choose another slot or date

Modification not possible

At step 11:

- a. System displays error (e.g., too close to booking time)
- b. User is returned to “My Bookings”

UC3: CANCEL A BOOKING

Use case: Cancel an existing parking booking

Primary actor: SETU User

Preconditions:

- User has already made a booking
- User has signed in

Main success scenario

1. User navigates to “My Bookings”
2. System displays list of current bookings
3. User selects the booking to cancel
4. System displays booking details and cancellation warning
5. User selects “Cancel Booking”
6. System prompts user to confirm cancellation
7. User confirms
8. System processes cancellation
9. System refunds the credits to the user’s weekly balance (if within policy)
10. System displays cancellation confirmation screen
11. System sends confirmation email

Alternative success scenarios

Non-refundable cancellation

- At step 9, system shows “No credits refunded (cancellation too close to date)”
- User confirms and proceeds

User changes mind

- User cancels the cancellation request and returns to “My Bookings”

Failure scenarios

Cancellation request failed

At step 8:

- a. System displays error
- b. User is returned to “My Bookings”

Booking reference invalid / already used

- System shows error and returns user to step 1

UC4: MAKE PAYMENT

Use case: Complete payment for a booking or modification

Primary actor: SETU User

Preconditions:

- User is in the process of booking or modifying a slot that requires extra credits/payment

Main success scenario

1. System displays payment summary (credits required or card payment)
2. User chooses to pay with saved card or enter new card details
3. System displays form: cardholder name, card number, expiry date, CVC, billing address
4. User fills out form and presses Submit
5. System authorises payment
6. System deducts credits or processes card payment
7. System returns to the booking/modification flow

Alternative success scenarios

Payment using credits only

- System automatically deducts from user's weekly credit balance (no card needed)

Refund / credit owed

- System displays credit message and automatically applies it

Failure scenarios

Card number invalid

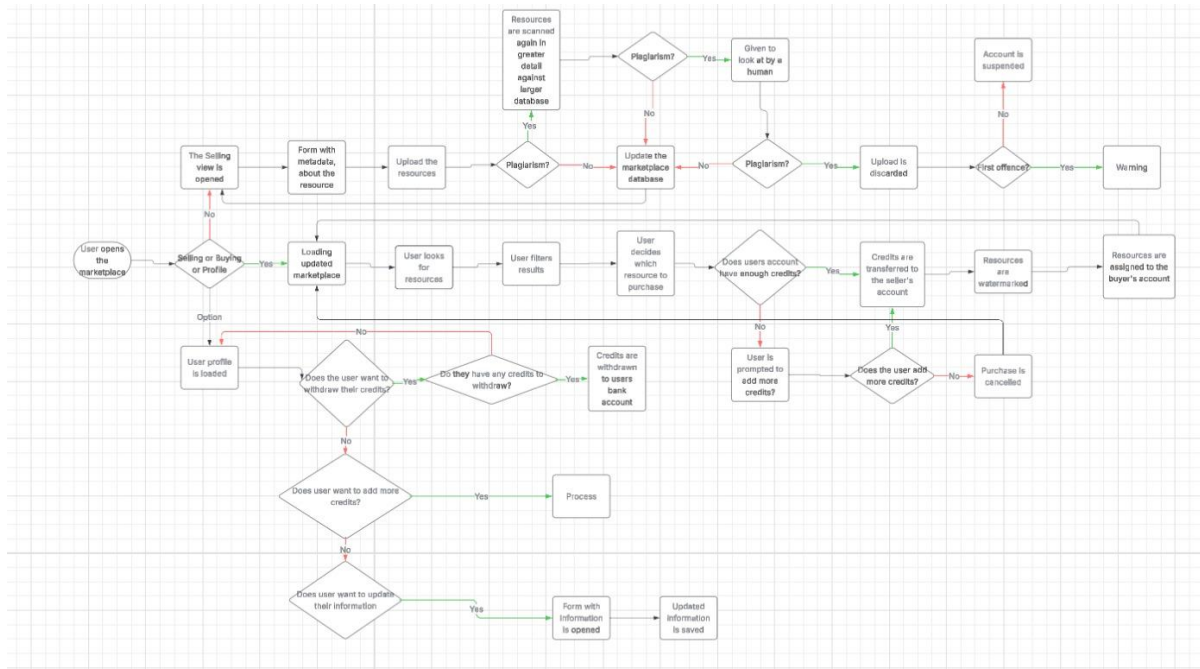
- System shows error and returns user to step 3

Payment authorisation failed

- System displays reason for failure
- User is returned to card details form to try again or use a different card

Marketplace Section:

Activity Diagram:



Personal Reflections

Bao:

What was your primary role and most significant contribution to the project?

I was an overseer. After I do my section, I check if my teammates' sections are satisfactory, then merge all the documents together to get the final document to submit. I would say my most significant contribution is not one thing, but just the overall planning and asking the lecturer for clarification, besides checking on progression. It helped keep the team on track and on schedule.

Describe one challenge your team faced and explain how it was addressed.

The most frequent challenge our team faced was time management. Although Trello did help to alleviate this at the start, once week 10 hit, there was just a lot to do at the same time. It will be much worse in weeks 11 and 12 though, because that is when every module start to end. To address this, we are accelerating our schedule and rearranging tasks, trying to finish it before week 11. So, one of my team members did a section for me, specifically the user diagram, while I took care of the test for him, playing to our strengths.

Identify one technical or process decision that affected the project outcome.

The decision to make our project three individual minor projects merged into one was the biggest decision that affected the outcome drastically. It was to guarantee everyone had a balanced amount of work to prevent piggybacking, whether accidental or on purpose. I'd say it was a rushed decision because, even though the benefits are clear, it meant we did a lot more work than was required.

What is one thing you would do differently in a future team software project, and why?

One thing I would change next time is to think about the task distribution more clearly. We would accept that it's hard for everyone to do the same amount of work, so just distribute the work to each member's strengths like the example I drew above. That, I believe, would be the optimal solution, and what we should've done from the start.

Gustaw:

What was your primary role and most significant contribution to the project?

Over the course of the assignment, I contributed to my group which I was in with Bao and Joey. I did my part while they did theirs. I contributed at all stages of the project where I finished my work and further on sent to Bao for him to merge it together with his and Joey's work. I am happy with my contribution and would say that each of us did a similar amount of work on this assignment.

Describe one challenge your team faced and explain how it was addressed.

For me the largest challenge was the formatting of different parts of the assignment. I had trouble finding some references or other things which would help me a lot with finishing my part in the format required by the lecturer. Other than that, I think all the tasks were straightforward and explained well in the course material on Moodle. While my teammates divided their tasks between them to manage their strengths and weaknesses for the most efficient completion of the assignment. I just did all the parts on my own since it was far easier for me to manage my own workflow within the assignment by following the course material on Moodle.

Identify one technical or process decision that affected the project outcome.

The project ended up being 3 smaller projects where we in the end merged to make a large final assignment. This helped manage the work efficiently and during the work phase we wouldn't be in each other's way when everyone worked on their thing. It also prevented from one teammate not contributing anything to the project and improved the equal division between tasks.

What is one thing you would do differently in a future team software project, and why?

Next time I would like to improve the communication about formatting of the parts that we do. I don't think it was clear enough causing more work during the merging process. It could've easily been avoided with better communication where we could've saved time in the final stages of each task.

Joseph:

What was your primary role and most significant contribution to the project?

Well, as the project consisted of three separate app entities within one all-encompassing app, obviously my primary role was to design and implement one of the three. Although that statement isn't entirely true. There were somewhat small alterations between the team's workflow that weighed certain work on top of one than other work. In my case, I was prescribed to focus more on the UML modelling, whereas previously the team, as per usual, would divide amongst one another an even split regardless. However, sometimes one member of a team is more readily willing to undertake tasks such as involving the J-Units tests, therefore in compensating, I readily took on more diagram related tasks.

Describe one challenge your team faced and explain how it was addressed.

Early into undertaking the project, we figured we had to settle on an application idea as what would naturally feel conventional. However, I felt that to have quickly moved on from that situation of indecision, we mutually decided to incorporate all of our respective app's ideas we were most fixated on, which wasn't difficult because our ideas (as assigned) were all in relation to SETU services. Otherwise, the conception would appear too lazy, but again due to the SETU context; we felt it blended well.

Identify one technical or process decision that affected the project outcome.

As our team ventured with our project, on account of our idea being to amalgamate our ideas into a cohesive whole, we had almost full liberty to experiment with how we developed our own personal corner of the app. Otherwise I'd predict, there'd be conflict between one another because if the app was solely based around an idea such as e.g. Prioritised Parking. We'd have to consult one another regularly to make sure any of our designs and implementations didn't conflict / contradict with one another. In summary, with our own ideas, the project workflow was always continuous with no, if you will, pauses due to one team member or another stepping outside the project vision, hindering each other and forestalling the projects realisation.

What is one thing you would do differently in a future team software project, and why?

Not that our team co-operation was weak, in fact, it was very successful in optimizing our project. However, perhaps in the future with the early stage of developing a project there'd be more discussion around a single idea. If we all had to put our minds together the outcome could only be very interesting as a concept. Overall, though when there's other modules to split your time and energy across, you can't dwell on one module for too long and you of course, you have to consistently be completely one stage and moving on to the next in a project within a certain time frame and also naturally its college work at the end of the day so it's the type of scenario where you want to be as efficient as possible so you have more time away from you work. But who knows maybe in a future workplace or otherwise when the circumstances give you free reign to gather up an intriguing idea and vision, could this project approach I'm curious about be granted.